

$$f(u_1, u_2, \dots, u_n) = F(u_1, u_2, \dots, u_n, 1).$$

Then this restriction allows to reconstruct whole F

$$F(u_1, u_2, \dots, u_n, z) = f\left(\frac{u_1}{z}, \dots, \frac{u_n}{z}\right).$$

Proof:

$$F(u_1, u_2, \dots, u_n, z) = F\left(z \cdot \frac{u_1}{z}, \dots, z \cdot \frac{u_n}{z}, z \cdot 1\right) = z^k F\left(\frac{u_1}{z}, \dots, \frac{u_n}{z}, 1\right) = f\left(\frac{u_1}{z}, \dots, \frac{u_n}{z}\right)$$

19.4.2 Homothetic Functions and Properties

Homotheticity is a generalization of homogeneity. In fact, we will show that homothetic functions are the ordinal equivalent of homogeneous functions.

Definition: A function $y: r^n \rightarrow R$ is called homothetic if it is a monotonic transformation of a homogeneous function, that is there exist a strictly increasing function $g: R \rightarrow R$ and a homogeneous function $x: R^n \rightarrow R$ such that $y = g \circ x$

Homotheticity is an ordinal property: monotonic transformation of homothetic function is homothetic.

Example 19.2 : Let $x(u, v) = uv$, a homogeneous function of degree 2. Then the monotonic transformations

$$g_1(z) = z + 1, g_2(z) = z^2 + z, g_3(z) = \ln z,$$

generate the following homothetic (but not homogeneous) functions

$$y_1(u, v) = uv + 1, y_2(u, v) = u^2v^2 + uv, y_3(u, v) = \ln u + \ln v.$$

Properties of Homothetic Functions:

Theorem D: Level sets of a homothetic function are radial expansions of one another, that is $y(u) = y(v)$ implies $y(tu) = y(tv)$ for arbitrary $u = (u_1, u_2, \dots, u_n), v = (v_1, v_2, \dots, v_n), t > 0$.

Proof: $y(tu) = g(x(tu)) = g(t^k x(u)) = g(t^k x(v)) = g(x(tv)) = y(tv)$.

Theorem E: For a homothetic function the slopes of level sets along rays from the origin are constant, that is

$$-\frac{\frac{\partial y}{\partial u_i}(tu)}{\frac{\partial y}{\partial u_j}(tu)} = -\frac{\frac{\partial y}{\partial u_i}(u)}{\frac{\partial y}{\partial u_j}(u)}$$

For all i, j and $t > 0$.

In other words, Marginal Rate of Substitution (MRS) for a homothetic function is a homogeneous function of degree 0.

Proof:

$$\begin{aligned} \frac{\frac{\partial y}{\partial u_i}(tu)}{\frac{\partial y}{\partial u_j}(tu)} &= \frac{\frac{\partial}{\partial u_i} g(x(tu))}{\frac{\partial}{\partial u_j} g(x(tu))} = \frac{g'(x(tu)) \cdot \frac{\partial x}{\partial u_i}(tu)}{g'(x(tu)) \cdot \frac{\partial x}{\partial u_j}(tu)} = \frac{\frac{\partial x}{\partial u_i}(tu)}{\frac{\partial x}{\partial u_j}(tu)} \\ &= \frac{t^{k-1} \frac{\partial x}{\partial u_i}(u)}{t^{k-1} \frac{\partial x}{\partial u_j}(u)} = \frac{\frac{\partial x}{\partial u_i}(u)}{\frac{\partial x}{\partial u_j}(u)}. \end{aligned}$$

Substituting $t = 1$ we get

$$\frac{\frac{\partial y}{\partial u_i}(u)}{\frac{\partial y}{\partial u_j}(u)} = \frac{\frac{\partial x}{\partial u_i}(u)}{\frac{\partial x}{\partial u_j}(u)},$$

Thus

$$-\frac{\frac{\partial y}{\partial u_i}(tu)}{\frac{\partial y}{\partial u_j}(tu)} = -\frac{\frac{\partial x}{\partial u_i}(u)}{\frac{\partial x}{\partial u_j}(u)} = -\frac{\frac{\partial y}{\partial u_i}(u)}{\frac{\partial y}{\partial u_j}(u)},$$

This completes the proof.

Theorem F: If x is homothetic and y is ordinally equivalent to x , then y is also homothetic.

Proof: Suppose $y(u) \geq y(v)$. Then $x(u) \geq x(v)$ by equivalence. For all $t > 0$, $x(tu) \geq x(tv)$ by homotheticity of x and then $y(tu) \geq y(tv)$ for all $t > 0$ by equivalence.

Check Your Progress 3

- 1) Let $f(u_1, u_2) = 3u_1^5 u_2 + 2u_1^2 u_2^4 - 3u_1^3 u_2^3$. Find out homogeneity and degrees of homogeneous function.

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- 2) Determine whether the function $f(u_1, u_2) = \alpha \ln(u_1) + \beta \ln(u_2)$ is homothetic.

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Let us recall that a function $f(u)$ is called concave in an interval $[u_1, u_2]$ (where $u_1 < u_2$) if the line joining any two points on the graph is never above the curve. Let function f be continuous in the interval X and twice differentiable in the interior of that interval, X^0 . Then,

f is concave on $X^0 \Leftrightarrow f''(u) \leq 0$ for all $u \in X^0$.

Example 19.3: Is $f(u) = au^2 + bu$ a concave function?

Ans. $f(u) = au^2 + bu$. Differentiating both sides w.r.t. u , $f'(u) = 2au + b$. Again, differentiating both sides w.r.t. u , $f''(u) = 2a$. Now, if $a < 0$, then $f''(u) < 0$. Then the function $f(u)$ would be concave.

19.5.1 Maximum and Minimum Condition for Concave Functions

Let $f(u)$ is a concave function in an interval X . If a is a stationary point in the interval X for $f(u)$, then a is a minimum (maximum) point for $f(u)$ in X .

In short, when a is an interior point of X , and the following conditions are satisfied:

- 1) $f''(u) \geq 0$ for all values of $u \in X$
- 2) $f'(a) = 0$

Then, this implies $u = a$ is a minimum point for $f(u)$ in X .

Similarly, when a is an interior point of X , and the following conditions are satisfied:

- 1) $f''(u) \leq 0$ for all values of $u \in X$
- 2) $f'(a) = 0$.

Then, this implies $u = a$ is a maximum point for $f(u)$ in X .

Formal Graphical Definition of Concave Functions

In the above section we have defined concavity only for twice differentiable function. There is a way to characterize concavity which is applicable even when the function is not differentiable and that is through a geometric (graphical) way. A function $f(u)$ is called concave if the line joining any two points on the graph is never above the curve (Fig. 19.1).

One can easily prove that for differentiable functions, the definition of concavity in terms of the sign of the second derivative is equivalent to the geometric definition of concavity. To examine the geometric definition of concavity, we must have an algebraic formulation of the geometric definition of the given function.

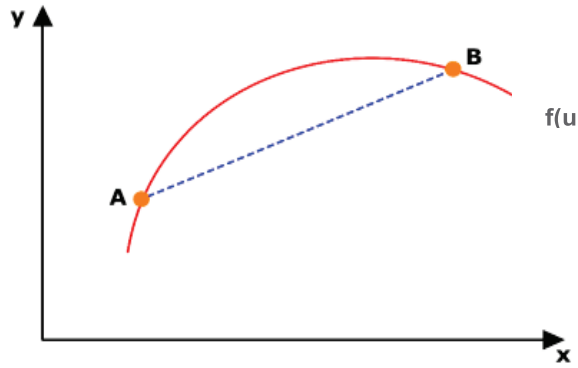


Fig. 19.1: Concave Function

Let there exist an arbitrary point u in the interval $[a, b]$ where $a < b$ and for some $\alpha \in [0, 1]$ can be written as

$$u = (1 - \alpha)a + \alpha b = a + \alpha(b - a).$$

If $a < b$ and $0 \leq \alpha \leq 1$, then $a \leq \alpha(b - a) \leq b$. Conversely, if $u \in [a, b]$ and we use $u = \frac{u - a}{b - a}$ then $0 \leq \alpha \leq 1$ and $(1 - \alpha)a + \alpha b = \left(1 - \frac{u - a}{b - a}\right)a + \frac{u - a}{b - a}b = u$.

Here α is the ratio between the distance from u to a and the total distance between a and b (Fig. 19.2).

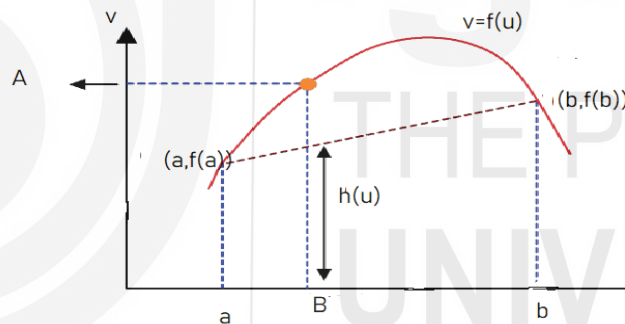


Fig. 19.2: Concave function with a line segment

where $A = f[(1 - \alpha)a + \alpha b]$ and $B = [(1 - \alpha)a + \alpha b]$

Now according to the definition of concavity, a function f would be concave if the graph of a function is on or above the line segment. So, a line segment joining $(a, f(a))$ and $(b, f(b))$ has the equation

$$v - f(a) = \frac{f(b) - f(a)}{b - a}(u - a)$$

Denote the height of the line segment from $(a, f(a))$ and $(b, f(b))$ at the point u by $h(u)$.

Thus, $v = h(u)$ and we put the value of u so,

$$h(u) - f(a) = \frac{f(b) - f(a)}{b - a}((1 - \alpha)a + \alpha b - a) = \alpha[f(b) - f(a)]$$

This implies that $h(u) = (1-\alpha)f(a) + \alpha f(b)$. Now as $\alpha \in [0,1]$ so the number $(1-\alpha)a + \alpha b$ can also take all the values in $[a,b]$.

Now, as the definition says that the line segment should always lie below or on the graph of f , this implies

$$h(u) \leq f((1-\alpha)a + \alpha b) \text{ for all } \alpha \in [0,1].$$

$$(1-\alpha)f(a) + \alpha f(b) \leq f((1-\alpha)a + \alpha b), \text{ for all } \alpha \in [0,1].$$

Definition: A function f defined on the convex set $X \subset \mathbb{R}^n$ is called concave if for every $u_1, u_2 \in X$ and $0 \leq t \leq 1$, we have

$$(1-\alpha)f(a) + \alpha f(b) \leq f((1-\alpha)a + \alpha b).$$

Concave Functions of One Variable:

Theorem G: Let f be a differentiable function on the open convex set $X \subset \mathbb{R}$. It is concave if and only if for every $u_o, u \in C$, we have

$$f(u) \leq f(u_o) + f'(u_o)(u - u_o).$$

Theorem H: Let f be a differentiable function on the open convex set $X \subset \mathbb{R}$. It is concave if and only if f' is a non-increasing (decreasing) function

Concave Functions of Several Variables:

Definition: For any function f on X and any $\lambda \in \mathbb{R}$ the set $U(f, \lambda)$ defined by

$$U(f, \lambda) = \{u : u \in X, f(u) \geq \lambda\}$$

is called the upper level (or contour) set of f . The set

$$L(f, \lambda) = \{u : u \in X, f(u) \leq \lambda\}$$

is called the lower level (or contour) set of f .

Proof: Let u' and u'' be two points in $U(f, \lambda)$, i.e., $f(u') \geq \lambda$ and $f(u'') \geq \lambda$. By concavity of f ,

$$f(\alpha u' + (1-\alpha)u'') \geq \alpha f(u') + (1-\alpha)f(u'') \geq \alpha \lambda + (1-\alpha)\lambda = \lambda.$$

Therefore, $\alpha u' + (1-\alpha)u'' \in U(f, \lambda)$.

The converse statement of the above theorem is not always true. To guarantee the sufficiency we would need a weaker condition called quasi concavity.

Theorem I: Let f be a differentiable function on the open convex set $X \subset \mathbb{R}^n$. It is concave if and only if for every $u_o, u \in X$, we have

$$f(u) - f(u_o) \leq \nabla f(u_o) \cdot (u - u_o)$$

Proof: Let us assume that f be concave on X . Let $u_o, u \in X$, by the concavity of f , we have $\forall \alpha \in [0,1]$,

$$f(u_o + \alpha(u - u_o)) \geq f(u_o) + \alpha[f(u) - f(u_o)]$$

$$\Rightarrow \frac{f(u_0 + \alpha(u - u_0)) - f(u_0)}{\alpha} \geq f(u) - f(u_0)$$

Taking the limit of the above expression as $\alpha \rightarrow 0^+$, we have

$$f(u) - f(u_0) \leq \nabla f(u_0) \cdot (u - u_0).$$

To prove the converse, take u' and u'' in X , and let $u^\alpha = \alpha u' + (1 - \alpha)u''$. By assumptions,

$$f(u') \leq f(u^\alpha) + \nabla f(u^\alpha) \cdot (u' - u^\alpha),$$

$$f(u'') \leq f(u^\alpha) + \nabla f(u^\alpha) \cdot (u'' - u^\alpha)$$

Multiplying the first inequality by α and the second inequality by $1 - \alpha$, and adding we get

$$f(u^\alpha) \geq \alpha f(u') + (1 - \alpha)f(u'').$$

Hence proved.

Quasi-Concave Functions

Theorem J: Let f be defined on the convex set $X \subset R^n$. It is a quasi-concave function if and only if

$$f(\alpha u_1 + (1 - \alpha)u_2) \geq \min[f(u_1), f(u_2)] \text{ for every } u_1, u_2 \in X \text{ and } 0 \leq \alpha \leq 1.$$

Proof: First suppose that $U(f, \alpha)$ is convex. Take $u_1, u_2 \in X$ and $\alpha \in [0, 1]$. Assume without loss of generality that $f(u_1) \geq f(u_2) = \alpha$. Thus, $u_1, u_2 \in U(f, \alpha)$, and by convexity of $U(f, \alpha)$ we have $\alpha u_1 + (1 - \alpha)u_2 \in U(f, \alpha)$, which means

$$f(\alpha u_1 + (1 - \alpha)u_2) \geq \alpha = f(u_2) = \min[f(u_1), f(u_2)].$$

To show the converse, suppose that $f(\alpha u_1 + (1 - \alpha)u_2) \geq \min[f(u_1), f(u_2)]$ for all $u_1, u_2 \in X$ and for all $\alpha \in [0, 1]$. If $U(f, \alpha)$ is empty or singleton, then it is trivially convex. So, suppose that $U(f, \alpha)$ contains at least two points u_1 and u_2 . Then $f(u_1) \geq \alpha$ and $f(u_2) \geq \alpha$, and so $\min[f(u_1), f(u_2)] \geq \alpha$. By hypothesis $f(\alpha u_1 + (1 - \alpha)u_2) \geq \min[f(u_1), f(u_2)]$ and hence $\alpha u_1 + (1 - \alpha)u_2 \in U(f, \alpha)$. Since α was chosen arbitrarily, this completes the proof of the first part.

Alternatively:

Definition: Let f be defined on the convex set $X \subset R^n$. It is said to be quasi concave if its upper-level sets $U(f, \alpha) = \{u : u \in X, f(u) \geq \alpha\}$ are convex sets for every real α .

Differentiable Quasi-Concave Functions

Quasi Concave functions have derivative characterizations as do the concave functions.

Theorem K (Quasi concavity for C^1 function is a continuous function with a continuous first derivative] functions)

Let f be differentiable on the open convex set $X \subset R^n$. Then f is quasi concave if and only if for every $u_1, u_2 \in X$

$$f(u_2) \geq f(u_1) \Rightarrow \nabla f(u_1) \cdot (u_2 - u_1) \geq 0$$

Proof: First, suppose that f is quasi concave on X , and let $u_1, u_2 \in X$ such that $f(u_2) \geq f(u_1)$. Let $\alpha \in [0,1]$. Since f is quasi concave, we have

$$\begin{aligned} f(u_1 + \alpha(u_2 - u_1)) &\geq \min[f(u_1), f(u_2)] = f(u_1) \\ \Rightarrow \frac{f(u_1 + \alpha(u_2 - u_1)) - f(u_1)}{\alpha} &\geq 0, \text{ for all } \alpha \in [0,1]. \end{aligned}$$

Taking the limit of the above expression as $\alpha \rightarrow 0^+$, we have $\nabla f(u_1) \cdot (u_2 - u_1) \geq 0$.

To show the converse, let us say $g(\alpha) = f(u_1 + \alpha(u_2 - u_1))$. Observe that $g(0) = f(u_1) \leq f(u_2) = g(1)$, and that g is a C^1 function on $[0,1]$ with the derivative $g'(\alpha) = \nabla f(u_1 + \alpha(u_2 - u_1)) \cdot (u_2 - u_1)$. Now suppose that, for all $u_1, u_2 \in X$, we have $f(u_2) \geq f(u_1) = \nabla f(u_1) \cdot (u_2 - u_1) \geq 0$. We will show that this implies the quasi concavity of f , i.e.,

$$g(\alpha) = f(u_1 + \alpha(u_2 - u_1)) \geq \min[f(u_1), f(u_2)] = f(u_1) = g(0), \text{ for all } \alpha \in [0,1]$$

Suppose on the contrary that the above is not true, i.e., there is some $\alpha_0 \in [0,1]$ such that $g(\alpha_0) < g(0)$. Because $g(0) \leq g(1)$, we can choose α_0 such that $g'(\alpha_0) > 0$. Now let $z_0 = u_1 + \alpha_0(u_2 - u_1)$. Because $f(u_1) = g(0) > g(\alpha_0) = f(z_0)$ and $z_0 \in X$, we have, by hypothesis, that

$$\nabla f(z_0) \cdot (u_1 - z_0) = \nabla f(z_0) \cdot (-\alpha_0)(u_2 - u_1) \geq 0 \Rightarrow \nabla f(z_0) \cdot (u_2 - u_1) \leq 0 \quad (19.27)$$

The above is true since $\alpha_0 > 0$. On the other hand, we have chosen α_0 such that

$$g'(\alpha_0) = \nabla f(z_0) \cdot (-\alpha_0)(u_2 - u_1) > 0 \quad \dots(19.28)$$

Inequalities (19.27) and (19.28) contradict each other. Therefore, there cannot be such $\alpha_0 \in [0,1]$ with $g(\alpha_0) < g(0)$, and we conclude that f is quasi concave.

Hence proved.

Theorem L

Let f be a twice differentiable quasi concave function on the open convex set $X \subset R^n$. If $u_o \in X$, $y \in R^n$, $y^T \nabla f(u_o) = 0 \Rightarrow y^T \nabla^2 f(u_o) y \leq 0$.

(VI) Determinant Criterion for Quasi-Concavity

First let us define the bordered Hessian determinants as the criterion for checking quasi-concavity of a function by checking the signs of certain determinants.

If $z = f(u_1, u_2)$ is twice differentiable with the continuous partial derivatives of the first and second order, defined in an open convex set X , then $z = f(u_1, u_2)$ is strictly quasi-concave if the bordered Hessian determinant be represented by $u_o \in X$,

$$D_b(u) = \begin{vmatrix} 0 & f'_1(u) & \cdots & f'_r(u) \\ f'_1(u) & f''_{11}(u) & \cdots & f''_{1r}(u) \\ \vdots & \vdots & \ddots & \vdots \\ f'_r(u) & f''_{r1}(u) & \cdots & f''_{rr}(u) \end{vmatrix} > 0$$

A necessary condition for f to be quasi-concave is that $(-1)^b D_b(u) \geq 0$ for $b = 1, \dots, n$ and all $u \in X$.

The sufficient condition for f to be quasi-concave is that $(-1)^b D_b(u) > 0$ for $b = 1, \dots, n$ and all $u \in X$.

Check Your Progress 4

- 1) Use the bordered Hessian condition to determine whether the function $f(u_1, u_2) = u_2 e^{-u_1}$ is quasi-concave for the region in which $u_1 \geq 0$ and $u_2 \geq 0$.

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- 2) Is the following function convex or concave? $f(u_1, u_2) = 2u_1^3 - 6u_2^2$

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19.6 ECONOMIC APPLICATIONS OF EULER'S THEOREM

Suppose the production function $q = f(u_1, u_2, \dots, u_n)$ is homogeneous of degree k and p be the price of one unit of the product. Then, by Euler's Theorem

$$u_1 p \frac{\partial f}{\partial u_1}(u) + \cdots + u_n p \frac{\partial f}{\partial u_n}(u) = kpq$$

Here, pq is the total value of output;

$\frac{\partial f}{\partial u_i}(u)$ is amount of additional output when the input u_i is increased by 1, i.e., the

marginal product of the i^{th} input;

$p \frac{\partial f}{\partial u_i}(u)$ is price of additional output when the input u_i is increased by 1, i.e., n

marginal revenue of the i -th input, assuming u is profit maximiser, it equals to marginal cost of the i^{th} input;

$u_i p \frac{\partial f}{\partial u_i}(u)$ is the total cost of the i^{th} input;

$u_1 p \frac{\partial f}{\partial u_1}(u) + \dots + u_n p \frac{\partial f}{\partial u_n}(u)$ is the total cost of production at profit maximiser.

Thus, from Euler's Theorem follows:

If $k = 1$ the firm spends whole revenue on input, so no profit in this case.

If $k > 1$, then the total payment exceeds the value of output.

If $0 < k < 1$, then the total payment is less than the value of output, so the firm makes a positive profit.

19.7 LET US SUM UP

In this unit, we emphasised on the basic theory of constrained optimisation, with its division into two main parts viz., constrained optimisation in the case of equality constraints and constrained optimisation in the case of inequality constraints. We considered two different techniques for solving these two different types of problems. In case of optimisation under equality constraints, we apply Lagrangian method to solve those problems, and in the latter case, that is, in the case for optimisation under inequality constraints, we apply the Kuhn-Tucker theory. We saw that this theory is different in many respects. First, the first order conditions of Lagrangian theory and the Kuhn-Tucker theory are different. Secondly, in Kuhn-Tucker theory there is no need to check the second order condition, while in case of Lagrangian case one must check it to ensure whether the optimum value (that we get from first order condition) is maximum or minimum. Finally, in Kuhn-Tucker theory there is a new set of conditions called “complementary slackness condition”, which does not exist

We started with the Envelope Theorem for unconstrained optimization problems and moved towards more general case for constrained optimization problems. We then presented characterizations of optimisation problems under a weakening of the condition of concave functions, called “quasi-concave” functions. Some of the characterizations of these functions as a generalization of concave functions and proves are discussed in terms of its contour sets (level sets), derivatives and extreme properties of the functions on line segment.

19.8 KEY WORDS

Bordered Hessian : Let f be a twice-differentiable function of n Variables. For $r=1, \dots, n$ the r th order Bordered Hessian of f at the point u is the matrix

$$\begin{pmatrix} 0 & f'_1(u) & \cdots & f'_r(u) \\ f'_1(u) & f''_{11}(u) & \cdots & f''_{1r}(u) \\ \vdots & \vdots & \ddots & \vdots \\ f'_r(u) & f''_{r1}(u) & \cdots & f''_{rr}(u) \end{pmatrix}$$

Cardinal Property of a Function : A property of a function is called cardinal if it depends on the shape, location of level sets and also the actual values of the function.

Euler's Theorem: : Let $f(u_1, u_2, \dots, u_n)$ be a function that is homogeneous of degree k . Then

$$u_1 \frac{\partial f}{\partial u_1}(u) + \cdots + u_n \frac{\partial f}{\partial u_n}(u) = kf(u)$$

or, in gradient notation $u \cdot \nabla f(u) = kf(u)$

Monotonic Transformation : Such integrals are evaluated using the concept of Let I be an interval on the real line. Then $g: I \rightarrow R$ is a monotonic transformation of I if g is strictly increasing function on I . Furthermore, if g is a monotonic transformation and u is a real valued function of n variables, then $g \circ u: x \rightarrow g(u(x))$ is a monotonic transformation of u .limits.

Ordinal Property of a Function : A property of a function is called ordinal if it depends only on the shape and location of level sets and does not depend on the actual values of the function.

19.9 ANSWERS/HINTS TO CHECK YOUR PROGRESS EXERCISES

Check your Progress 1

1) $L = u_1 u_2 + \lambda(100 - u_1 - u_2)$

Now take partial derivatives, one for each unknown, including λ

$$\frac{\partial L}{\partial u_1} = u_2 - \lambda = 0$$

$$\frac{\partial L}{\partial u_2} = u_1 - \lambda = 0$$